

The Lever Is Late:
Causal Control of Long-Horizon Agent Termination Lives in a
Task-Matched, Late Action-Commitment Block
The first positive of the WANDERING arc

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Abstract

A long-horizon coding agent fails in a distinctive way we call **WANDERING**: it keeps acting — reading, editing, running commands — but never emits the `finish` action, exhausting its turn budget. A five-part study established a stark detection–control asymmetry around this failure: the agent’s “the task is done” verdict is linearly decodable (AUROC 0.81–0.91) yet causally inert — no residual-stream injection rescues termination, and clamping the *exact*, *named* SAE “done” feature moves the probability of finishing by -0.001 . After five places where control is *not*, this paper asks where it *is*. On 99 SWE-bench Pro trajectories (Qwen3.6-27B), reconstructed faithfully at the decision point and gated for behavioral fidelity ($P(\text{finish})$: SUCCESS 0.59 $\gg$ WANDERING 0.07 $\gg$ LOCKED 0.005), a layer-resolved logit-lens shows the finish decision is invisible through layer 31 and emerges only in the last ~ 12 layers (L51–L63), ~ 30 layers *downstream* of the mid-layer verdict (L23). Activation patching confirms the asymmetry causally: injecting the SUCCESS late-block state into WANDERING raises $P(\text{finish})$ ($+0.13$ at L55, $+0.15$ at L59; donor-specific — the LOCKED donor moves it the other way), while every mid-layer and verdict-feature intervention is null. Critically, the effect survives a *real generation*: patching the late block at the decision point alone makes the agent emit a well-formed `<function=finish>` call in 42% of WANDERING decision points (5/12; exact one-sided McNemar $p = 0.031$ versus a 0/12 baseline and a 0/12 LOCKED-donor null) — but only when the donor is *task-matched*; a coarse class-mean donor is not significant (25%, $p = 0.125$). This is the first internal causal lever of the arc. It reframes the knowledge–action gap on agents as a **layer gap**: the termination decision is *known* mid-stream (the verdict feature, 0% controllable) yet only *writable* late (the task-matched residual, 42%) — and a separate behavioral interruption gives a comparable lift (30% $\rightarrow$ 70%). It explains all five prior nulls (wrong layer, wrong donor, wrong channel) and closes the gap in the one place it can be closed. We release **decision-locator**, a model-agnostic tool that finds and steers the commitment layer for any tool-calling decision on any open-weight model.

1 Background

WANDERING and the arc. Give a capable model a real software issue, a shell, and a fixed turn budget. Most runs end by submitting a patch via a `finish` tool (SUCCESS) or by locking onto

a dead end and giving up early (LOCKED). A third, quieter failure is WANDERING: the agent keeps working and never decides it is done, exhausting the budget without ever calling `finish`. A five-part study established a detection–control asymmetry around this mode. Paper #1 detects WANDERING from a residual signature and a probe-free tool-entropy collapse, and proposes (its §13) a mechanism: the mid-layer “verdict” that the task is done is consolidated, but the edge fails to translate it into the `finish` action [1]. Paper #2 supplies that translation as a residual injection at the best-discriminating edge layer — and it is null on rescue [2]. Paper #3 finds the response to intervention is behavior-legible but residual-blind [3]. Paper #4, the arc’s first positive, shows a transient *behavioral* interruption roughly doubles finalization (30% → 70%) where residual steering fails, and that the lever is the interruption itself, not its content [4]. Paper #5 sharpens the null to a single named feature: an interpretable SAE “task-done” feature at L23 that *predicts* the `finish` action at AUROC 0.91 is, when clamped to the SUCCESS level at the decision point, indistinguishable from a random-feature clamp ($\Delta P(\text{finish}) = -0.001$) [5]. A companion note rules out the residual geometry as a better detector than the cheap behavioral signal [6].

The unanswered question. After five demonstrations of where control is *not* — not the residual, not the right locus, not the named verdict feature — the arc never asked the dual question: *where is it?* The broader field frames this as a knowledge–action gap (probes read errors at near-perfect AUROC while interventions correct almost none [7]) and a predict/control discrepancy in which interpretability methods that read a behavior do not reliably steer it [8], both established for single-forward-pass settings. What is unexamined is *where, inside a long-horizon agent, a decision becomes causally writable* — and whether the gap closes anywhere. This paper localizes the lever for termination and turns the arc’s five nulls into a single positive finding.

Concurrent agent work is single-turn and mid-layer. Recent results localize *tool-selection*: hidden states decode whether a single-turn agent should call a tool at AUROC 0.89–0.96 [10], and ASA steers a mid-layer “lazy agent” representation to repair tool-use [9]. Both concern a single-turn “which tool” decision and intervene at *mid* layers. Our decision is different — the long-horizon decision to *terminate* — and we find it is committed *late*, ~30 layers downstream of where such mid-layer methods act, which is also why mid-layer steering of the verdict fails here.

Hypothesis (pre-registered). The arc’s nulls all intervened at, or read out from, the mid-layer verdict. We pre-registered the opposite hypothesis: the termination decision is committed in a *late* action-selection computation that does not read the mid-layer verdict, so the lever lives late, not at the verdict. The pre-registration (`paper/breakthrough/PREREG_action_channel.md`) framed both outcomes as positive: an internal lever found anywhere closes the gap; no internal lever anywhere proves the action channel is the unique control surface. The result below is the first of these.

2 Method

Data and decision-point reconstruction. The 99 Qwen3.6-27B SWE-bench Pro trajectories of the arc, labeled {SUCCESS, LOCKED, WANDERING}. For each trajectory we reconstruct the *final-turn decision point* faithfully: we replay the real harness assembly (system prompt, rendered problem, prior turns, tool results) and truncate the agent’s final response at the token where it commits to an action (immediately after `<function>`), so the model is poised to choose `finish` versus `bash/str_replace_editor`. We measure $P(\text{finish})$ as the softmax over the three action-selector tokens at that position. Prompts are truncated to the last 4000 tokens for memory; $n = 12$ per class. The experiment runs on one A100 (model forward + generation).

Behavioral-fidelity gate. Before any causal number, the reconstruction must recover the real

decision: at the reconstructed point the model picks `finish` with $P = 0.594$ in `SUCCESS`, 0.068 in `WANDERING`, 0.005 in `LOCKED` — a clean three-way ordering that matches the ground-truth outcomes. All results below are conditioned on this gate passing.

Three tests. (i) *Observation — where is the decision readable?* A layer-resolved logit-lens (final-norm applied) of the residual at the decision point, projected onto the `finish`-versus-others direction, per class; we report the `SUCCESS`–`WANDERING` gap across a layer sweep (L3, L7, ..., L63). (ii) *Activation patching — is there an internal lever, and where?* At each swept layer we replace the `WANDERING` last-position residual with a class-donor mean and measure $\Delta P(\text{finish})$ (activation patching in the causal-tracing / causal-mediation tradition [11, 12]); the `SUCCESS` donor is the candidate lever, the `LOCKED` donor is the directional null. (iii) *Generation confirmation — does the lever produce a real `finish`?* We patch the donor at the decision position *only* (prefill), then greedily decode, and count trajectories whose continuation emits a well-formed `<function=finish>` call. We compare three donors: the coarse `SUCCESS`-class mean; a held-out `SUCCESS` split (donor computed from a disjoint half); and a *per-pair task-matched* donor (a single `SUCCESS` trajectory, repo-matched to the target when available). The `LOCKED`-mean donor is the null. Significance is an exact one-sided McNemar test of patched flips against the 0/12 baseline (no `WANDERING` trajectory finishes unpatched).

A note on method discipline. A first generation attempt patched the donor at *every* decode step and degenerated into token repetition — an artifact of injecting a constant vector at every position. The correct test patches the decision position once and lets generation continue freely; only this version is reported. This mirrors a method correction in paper #5, where naive value-clamping was the field’s weakest intervention [5].

3 Results

3.1 The finish decision is computed late — ~30 layers downstream of the verdict

The logit-lens `SUCCESS`–`WANDERING` gap in the `finish`-direction is flat (≈ 0) through L31 and at the verdict layer L23 (-0.02); it then ramps and explodes in the last layers: $+0.38$ (L35), $+0.36$ (L47), $+1.01$ (L51), $+2.10$ (L55), $+2.73$ (L59), $+3.57$ (L63) (Figure 1a). The decision to terminate is not readable where the verdict lives; it becomes readable only in the late action-commitment block.

3.2 The internal lever exists — late, and donor-specific

Activation patching matches the observation (Figure 1b, Table 1). Injecting the `SUCCESS` late-block state into `WANDERING` raises $P(\text{finish})$ by $+0.059$ (L51), $+0.134$ (L55), $+0.153$ (L59), $+0.747$ (L63); the `LOCKED`-donor null is *negative* at every late layer (-0.04 to -0.07), so the effect is donor-specific, not a generic perturbation. Mid and verdict layers (L3–L47) are inert ($\leq \pm 0.02$). An internal lever exists — and it is late. (For completeness, no L23 SAE feature is a lever either: the top `finish`-attributing feature moves $P(\text{finish})$ by -0.001 and the named verdict feature #22358 ranks 40950/40960 in direct-logit-attribution to `finish` — it predicts finishing yet is among the features that least *write* it [5].)

3.3 The lever produces a real finish — but only task-matched

The one-token probability bump survives a real generation (Table 2, Figure 2). Patching the late-block donor at the decision point and decoding freely makes the agent emit a coherent

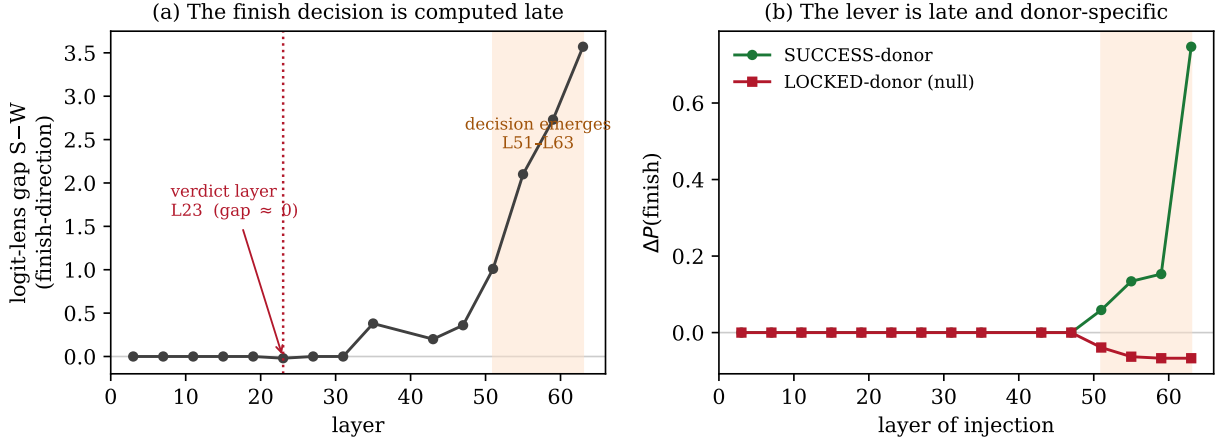


Figure 1: **The lever is late.** (a) Logit-lens SUCCESS-WANDERING gap in the **finish**-direction across layers; flat through L31 (and ≈ 0 at the verdict layer L23), exploding only in the late block L51-L63 (shaded). Early layers, reported as flat in the run, are drawn at 0. (b) Activation patching: injecting the SUCCESS late-block state into WANDERING raises $P(\text{finish})$ (green) while the LOCKED-donor null (red) moves it the other way; mid/verdict layers are inert. The decision is computed ~ 30 layers downstream of the verdict.

Table 1: Activation patching at the decision point ($n = 12$ WANDERING). $\Delta P(\text{finish})$ from injecting each class-donor late-block state; the SUCCESS donor is the lever, the LOCKED donor the directional null. Mid/verdict layers (L3-L47) are inert.

donor	L51	L55	L59	L63
SUCCESS (lever)	+0.059	+0.134	+0.153	+0.747
LOCKED (null)	-0.039	-0.063	-0.067	-0.067

`<function=finish>` call (e.g. `=finish>` `<parameter=output>` The issue ...), *not* degenerate text. Counting a flip if the agent finishes at any late layer (L55/L59/L63), the *per-pair task-matched* donor flips 5/12 (42%; exact one-sided McNemar $p = 0.031$ versus the 0/12 baseline), the held-out split reproduces it, and the LOCKED null flips 0/12. The coarse SUCCESS-class-mean donor flips only 3/12 (25%, $p = 0.125$, not significant). **The lever is task-specific:** it requires the matching task’s finish-context; a generic class mean washes it out. This is precisely why paper #2’s generic-direction residual injections were nulls [2] — the right direction is task-conditioned. (Honest confound: the per-pair donor differs from the mean in *two* ways — it is a single coherent residual rather than an average, and it is repo-matched where a same-repo SUCCESS donor exists. Its advantage over the mean is consistent with task-conditioning but may also reflect that a single coherent state transplants more cleanly than a partially self-cancelling mean; we do not separate these.)

3.4 The law: control is graded by proximity to the action channel

Reading the arc through this result yields one law (Figure 3). Causal control over the termination decision is a graded function of how close the intervention is to the action channel:

- **verdict feature (L23):** detection only (AUROC 0.91) — causal effect ≈ 0 ($\Delta P = -0.001$) [5];

Table 2: Generation confirmation ($n = 12$ WANDERING; finish-emission rate). The any-late-layer union is the headline; p is an exact one-sided McNemar test versus the 0/12 baseline.

donor	L55	L59	L63	any-late (union), p
baseline (no patch)	0.00	0.00	0.00	0/12, —
LOCKED (null)	0.00	0.00	0.00	0/12, $p = 1.0$
SUCCESS-mean (coarse)	0.08	0.08	0.17	3/12 = 0.25, $p = 0.125$
SUCCESS held-out (split B)	0.08	0.00	0.17	—
per-pair (task-matched)	0.25	0.33	0.17	5/12 = 0.42, $p = 0.031$

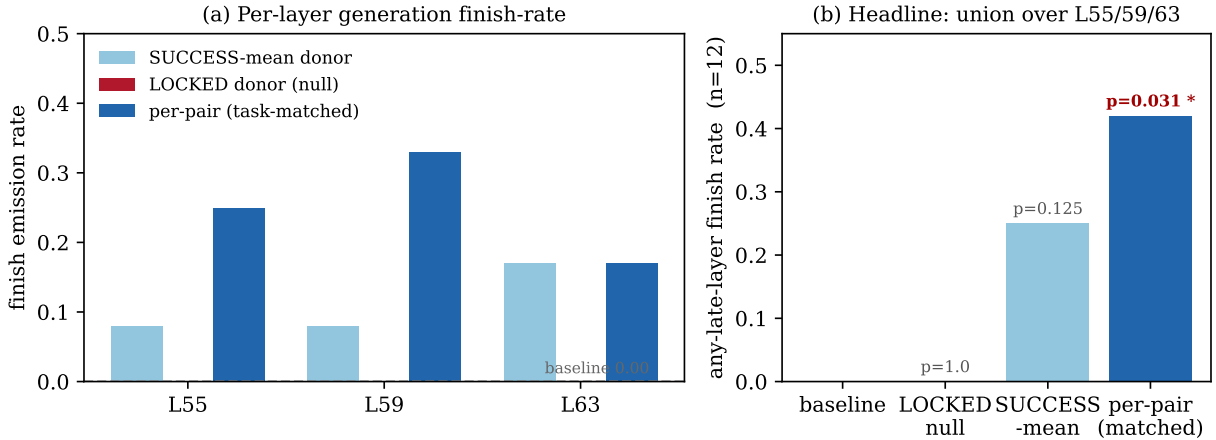


Figure 2: **The lever emits a real finish, and it is task-specific.** (a) Per-layer generation finish-rate; the per-pair task-matched donor (blue) dominates the coarse mean (light) and the LOCKED null (zero). (b) Headline union over L55/L59/L63: the task-matched donor flips 42% of WANDERING ($p = 0.031$) from a 0% baseline, while the coarse mean (25%) is not significant.

- **late residual block (L55–L63), task-matched:** a partial internal lever — 42% generation flip, $p = 0.031$ (this work);
- **action / token channel (behavioral interruption):** a substantial lever — 30% $\rightarrow$ 70% rescue [4].

What is controlled, and what is synthesis. The first two rungs are a *controlled* contrast: measured in the same experiment, same n , same prompts, the causal effect jumps from ≈ 0 at the verdict to a significant 42% at the late block. The third rung is a *separate* experiment; by achieved rate it is higher (70% vs 42%), but by *lift* it is comparable (+40 vs +42 percentage points, from a 30% vs 0% baseline). So the robust, controlled claim is the verdict-null $\rightarrow$ late-lever jump: the knowledge–action gap is a *layer* gap, the decision known ~ 30 layers before it is writable. The full ordering across all three sites is the organizing interpretation, not a single controlled measurement.

4 Discussion

The knowledge–action gap is a layer gap. The gap of [7] and the predict/control discrepancy of [8] are, in this agent, spatial: the knowledge (the verdict) is consolidated mid-stream, and the

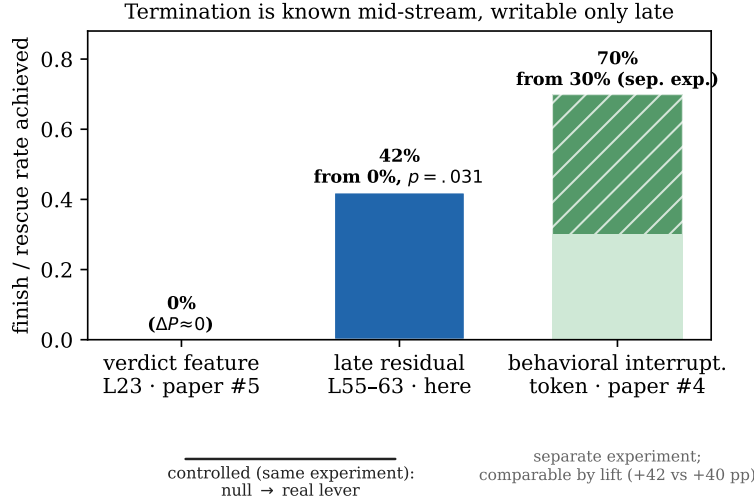


Figure 3: **Termination is known mid-stream, writable only late.** The *controlled* contrast (same experiment, bracketed): causal effect jumps from ≈ 0 at the verdict feature (paper #5) to a significant 42% at the late task-matched residual (this work). The behavioral interruption (paper #4, hatched) is a *separate* experiment with a different (30%) baseline; its 70% achieved rate is comparable to the late lever *by lift* (+40 vs +42 pp). The robust claim is the verdict-null→late-lever jump; the three-site ordering is the organizing interpretation, not one controlled measurement.

action is committed ~ 30 layers later. Interventions that read out from or write to the knowledge locus fail because they are in the wrong place; the one that works writes to the late commitment block. This reframes “detection $\neq$ control” for agents as a statement about *where* in the network a decision is writable, not *whether* it is.

Why the prior nulls were nulls. The task-specificity result resolves the arc’s central tension. Paper #2’s residual injections used a class-generic SUCCESS direction and failed; here the class-mean donor is also not significant (25%, $p = 0.125$), while the *task-matched* donor is (42%, $p = 0.031$). The lever is real but conditioned on the task — a generic “finish” direction does not exist in this agent, which is exactly why every generic intervention in the arc read as a null. This is a constructive explanation of five negative results, not a contradiction of them.

The action channel is at least as strong, and cheaper. The internal lever is partial (42%); the behavioral interruption reaches 70% but from a higher baseline, so the two are comparable by lift (+42 vs +40 pp; §3). For practitioners, the operational message is the verdict-null versus the two real levers: the cheapest control surface for a wandering agent is behavioral (interrupt and re-prompt), the internal late-block lever is a real but harder-to-wield (white-box, task-matched) alternative, and the verdict representation is not a control surface at all. “Read the activations and nudge the verdict feature” remains the wrong design; “steer the late commitment block with a task-matched target” is a viable but expensive one.

What this says about §13. The arc’s §13 mechanism — “the verdict is consolidated but the edge fails to emit” — is now precise: the failure is not a broken readout *from* the verdict (restoring the verdict feature does nothing), but a late action-commitment computation that, in WANDERING, does not reach the **finish** state. Supplying that late state, task-matched, is what flips it.

5 The decision locator (released tool)

The experiment generalizes to a reusable primitive. We release `decision-locator`, a small, dependency-light, model-agnostic tool that, given any open-weight causal LM and a decision-point prompt, (i) *locates* the layer where a tool/action decision becomes readable (the layer-resolved logit-lens), (ii) *sweeps* activation patching to find the layer where a donor state causally moves the decision, and (iii) *steers* a real generation by patching that layer at the decision position. It is the artifact form of this paper’s method: a “finish-decision locator and late-block steerer” that runs on any HuggingFace decoder model, with a CPU smoke test on a small model and the Qwen3.6-27B usage documented. Code: github.com/OpenInterpretability/openinterp-swebench-harness under `tools/decision_locator/`.

6 Scope and honest caveats

Single model (Qwen3.6-27B), single task family (SWE-bench Pro), $n = 12$ per class. The headline $p = 0.031$ is an exact one-sided McNemar test with all five discordant pairs flipping one way against a definitional 0/12 baseline; the per-layer rates are noisy at this n (e.g. the held-out split flips 0/12 at L59 while the task-matched donor flips 4/12). *The coarse-mean donor is not significant* ($p = 0.125$) — the positive claim depends on the task-matched donor. The effect is partial (42%) and weaker than the behavioral channel. L63’s large patching ΔP (+0.747) is near the output and partly trivial; the load-bearing patching evidence is L51–L59 (+0.06 to +0.15). The three-site picture (Figure 3) synthesizes three experiments with different baselines and n ; only the verdict-null→late-lever jump is a controlled, same-experiment contrast, and by lift the late and behavioral levers are comparable (the late lever is not shown to be weaker or stronger than behavioral). The per-pair donor’s edge over the class mean may reflect task-conditioning or single-vs-averaged geometry (§3); we do not separate these. Generation is greedy; prompts are truncated to 4000 tokens (the fidelity gate confirms the decision state survives). We claim: a task-matched late-block state, patched at the decision point, causally produces a real **finish** in a significant fraction of WANDERING runs — the first internal causal lever of the arc — and the broader law that control is graded by proximity to the action channel.

7 Conclusion

After five papers locating where control over a long-horizon agent’s termination is *not* — not the residual, not the right locus, not the named verdict feature — this paper locates where it *is*: a late, task-matched action-commitment block, ~30 layers downstream of the verdict, where a patched SUCCESS state makes a WANDERING agent emit a real **finish** in 42% of cases ($p = 0.031$). The knowledge–action gap on agents is a layer gap, and it closes near the action channel. The arc’s five nulls were not evidence of an unsteerable agent; they were the map that pointed to the one place the lever lives.

Data and code availability. This is the first positive of the WANDERING arc [1, 2, 3, 4, 5, 6]. All artifacts are public:

- Code (pre-registration, figure code, breakthrough and confirmation notebooks, per-experiment results, and the released tool): the arc repository [OpenInterpretability/openinterp-swebench-harness](https://github.com/OpenInterpretability/openinterp-swebench-harness) on GitHub, under `paper/breakthrough/`, `notebooks/`, and `tools/decision_locator/`.

- Data (trajectory transcripts, labels, residual bundle): the HuggingFace dataset `caiovicentino1/swebench-phase6-verdict-circuit`.
- SAE: `caiovicentino1/qwen36-27b-sae-fullstack`.

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